Probabilistic Machine Learning

Lecture 01: Introduction

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Speaker notes

Reworked 2026-09-08 along docs/reviews/PML-L01-REWORK-2026-09.md: the lecture is a guided progression prediction → uncertainty → its uses → its sources → where it enters a learning system → probability as the language → what the course enables → the road ahead (one ideal, several shortcuts). Formal vocabulary is deferred to the final section on purpose.

2026-09-24: students have taken machine-learning courses before, so the former "How Machines Learn" module (paradigms, programming vs learning, T/E/P, the y/α/h notation, the outer loop, design questions) is reduced to a one-slide recap of the learning system, read for uncertainty. The removed units remain in the library (the Data Science course uses them) and can be re-included.

2026-09-24: "Beyond True or False" uses the PML unit reasoning-cycle-under-uncertainty instead of ds/basics/v-modell: the reasoning cycle on Uranus/Neptune and Mercury/Vulcan, the choice between saving and changing a theory as a decision under uncertainty, and a fraud-model exercise (noise, too little data, or the wrong model?). The app/churn exercise stays in the Data Science course only.

TODO (needs input): the title slide should carry the University of Passau logo and the intended affiliation line (ITU fellowship). Neither asset nor wording exists in this repository — add the logo to figures/ and the wording here.

Why Probability?

Guiding Question: Two models both say "fraud". What would you need to know before acting on either — and why does a label not tell you?

Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

Café Central €4.20 legitimate	Supermarket €63.10 legitimate	**Electronics Online** €1,299.00 **fraud**	Petrol Station €78.40 legitimate
Gift Cards ×12 €600.00 **fraud**	Bookshop €24.90 legitimate	**Hotel Booking** €410.00 **fraud**	Pharmacy €12.75 legitimate

Speaker notes

Stepped visual, then discussion stepper. Click 1: the model stamps every transaction — five legitimate, three fraud. Click 2: the legitimate ones recede and the three flagged cases stand alone. Pause here: a €1,299 electronics order, twelve gift cards, and a hotel booking are obviously *different situations*, yet the model's output for them is one identical word.

Then reveal the questions one at a time and let the room answer before the next appears. Students should feel that a label gives them nothing to separate a near-certain case from a coin flip. Do not name the missing element — the next slide lets them discover it.

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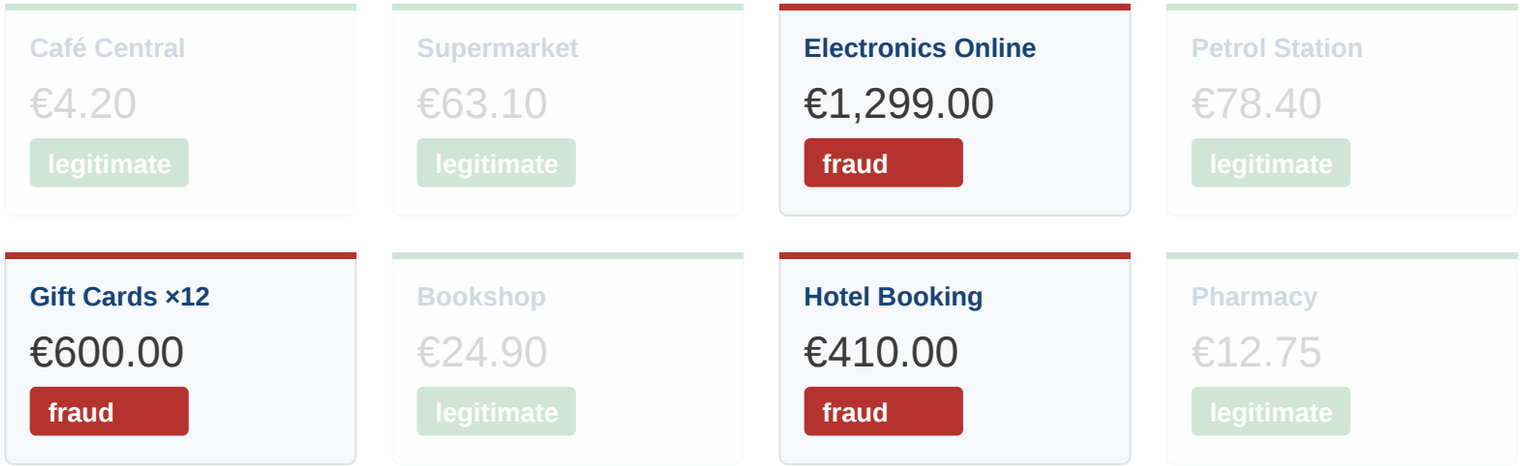
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Are the three flagged transactions equally suspicious?

Machine Learning Gives Us Predictions

Three are flagged as fraud

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Are the three flagged transactions equally suspicious?

Would you block all three cards?

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Are the three flagged transactions equally suspicious?

Would you block all three cards?

What would you need to know to treat them differently?



The Same Batch, With a Number Attached



What can you do now that you could not do a minute ago?

Speaker notes

Identical picture, identical batch — click 1 labels the batch again, click 2 adds the number and the bar, and the three cases that were indistinguishable separate: 0.98 against 0.55 and 0.51. Let students list what they can now do before revealing the cards (P2). Typical answers: rank them, act on the electronics order, send the other two to a person, ask why the model is unsure. Every answer is an action a bare label could not support — the number is not a nicer prediction, it is a **different kind of information**.

If someone asks whether 0.55 "is" a probability: only if the model is *calibrated* — of all cases it scores 0.55, about 55% really are fraud. Many scores are not. Lecture 04 shows when a model's output earns the name; for now, assume it does.

What a Number Lets Us Do

The number lets us	What that means for this batch
Compare and rank	separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud"
Weigh consequences	set the threshold from the cost of a missed fraud against the cost of an annoyed customer
Defer deliberately	send 0.55 and 0.51 to a human; ask whether more data would move them
Combine and communicate	merge evidence from several signals; report *how sure* we are, not only what we think

Speaker notes

The four cards generalise the room's answers; the reveal is the module's central message and should be spoken, not read. Both halves matter: representing uncertainty is the easy part, doing something principled with it is the course.

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> **We need to model uncertainty** — and we need to be able to **reason with it**.

From Prediction to Managed Uncertainty

Predict
a label or a value

→

Quantify
how sure we are

→

Model
represent it explicitly

→

Reason
combine, compare, update

→

Manage
act, defer, or gather data

And once we want to manage it — **where does the uncertainty actually come from?**

> Probabilistic machine learning is not a different way of producing predictions. It is a **language for reasoning about predictions that are not certain**.

Speaker notes

Bridge slide. Read the five cards as one sentence, left to right. The last card is the transition: before we can manage uncertainty we need to know what causes it — and the answer turns out to matter for what we can do about it.

Mini-Exercise

A loan applicant has a **default probability of 0.2**. A default costs the bank **€9,000**; rejecting a customer who would have paid costs **€1,000** in lost margin.

Accept or reject?

Speaker notes

One minute to read, one to think. Most students will say "accept, 0.2 is low" — let that answer stand for a moment before the click. The point is not the arithmetic but the reversal: a probability of 0.2 is high when the loss it guards against is nine times larger than the alternative.

Mini-Exercise

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Accept or reject?

Accepting costs $0.2 \cdot 9{,}000 = €1{,}800$ in expectation; rejecting costs $0.8 \cdot 1{,}000 = €800$. **Reject —** although p is far below 0.5.

What the Threshold Tells Us

THE INTUITIVE RULE

reject if $p > 0.5$

accepts this applicant — whose default risk is **twice** the bank's break-even

THE RULE THE COSTS IMPLY

reject if $p > 0.1$

because a default costs **9×** what a lost customer does: $p^* = \frac{1{,}000}{1{,}000+9{,}000}$

The threshold comes from the **costs**, not from the probability. 0.5 is only the right cut-off when both mistakes are equally expensive — which they almost never are.

Speaker notes

This slide answers the question students actually have — "where does the threshold come from?" — and corrects the intuition that 0.5 is neutral. At p* = 0.1 both expected costs are €900: that applicant is a genuine coin toss, and the place where a second signal is worth buying. Ask what happens to p* if the bank's margin doubles (€2,000: p* rises to 0.18, so it accepts more risk), and if the loan is ten times larger (€90,000: p* falls to about 0.011). The model output never changed; only the costs did.

Takeaway

A prediction without a measure of its uncertainty cannot be compared, weighed against consequences, or handed to a human at the right moment. **Modelling uncertainty and reasoning with it** is what turns a prediction into a decision.

All of this assumes the numbers are **calibrated**: a score of 0.55 is right about 55% of the time. Whether a model's output deserves that reading is a question Lecture 04 answers.

Where Uncertainty Comes From

Guiding Question: The model says 0.55. Is that because the world is noisy, or because the model has not learned enough — and would more data help?

Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

Speaker notes

Discussion stepper. Hold the two possibilities apart on the board: "the world is noisy" versus "we have not learned enough". Students usually reach the reducible/irreducible distinction themselves once the third question lands.

Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

Is the transaction itself ambiguous — would a perfect model, with all the data in the world, still be unsure?

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Or has the model simply not seen enough cases like this one?

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Is the transaction itself ambiguous — would a perfect model, with all the data in the world, still be unsure?

Or has the model simply not seen enough cases like this one?

Would collecting more data help — and could *any* amount of data remove the doubt entirely?

Two Examples

EXAMPLE A · AN AMBIGUOUS OBSERVATION

Two patients with **identical** records — same age, same symptoms, same test results. One recovers, one does not.

✗ Would more patients like these help? Never.

EXAMPLE B · AN UNFAMILIAR CASE

A house-price model trained on flats of 40–120 m² is asked about a **600 m² villa**. It has no examples anywhere near it.

✓ Would ten villas in the training data help? Completely.

Which uncertainty can be reduced, and which is part of the problem itself?

Speaker notes

Let students classify before the next slide. Push on Example A: "would a better model help?" — no, unless it sees *more attributes*; the noise is relative to the information available. That nuance is the honest version of "inherent".

The Distinction — Then the Names

	Where it comes from	More data?	Our example
Uncertainty in the problem	the outcome genuinely varies given what we observe	✗ never	identical patients, different outcomes
Uncertainty in our knowledge	we have not seen enough to pin the model down	✓ yes	the 600 m² villa

Speaker notes

The names arrive last, on a click, after the distinction has been made in plain words (intuition before terminology). Students who can only recite the Greek have learned nothing; students who can sort new cases into the two columns have learned the concept.

The Distinction — Then the Names

	Where it comes from	More data?	Our example
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Aleatoric

from alea, *dice* — noise in the world

Epistemic

from episteme, *knowledge* — ignorance in the model

Why the Distinction Matters

If the uncertainty is…	The right response	The expensive mistake
in our knowledge (epistemic)	collect data where the model is thin; defer to a human who knows more	accepting it as inherent and never improving
in the problem (aleatoric)	model the noise; set a policy for the ambiguous cases	buying more data that cannot help

A model that can tell the two apart can report that it is **out of its depth**, not merely unsure.

> A probabilistic model can represent **both** kinds of uncertainty. A Bayesian one can even report them separately — noise and ignorance as two terms of one prediction (Lecture 07). That is why the distinction is worth a name.

Takeaway

> Uncertainty has two sources: **noise in the world** and **gaps in our knowledge**. Only the second shrinks with data, and a decision — collect more, defer, act — depends on knowing which one you face.

Speaker notes

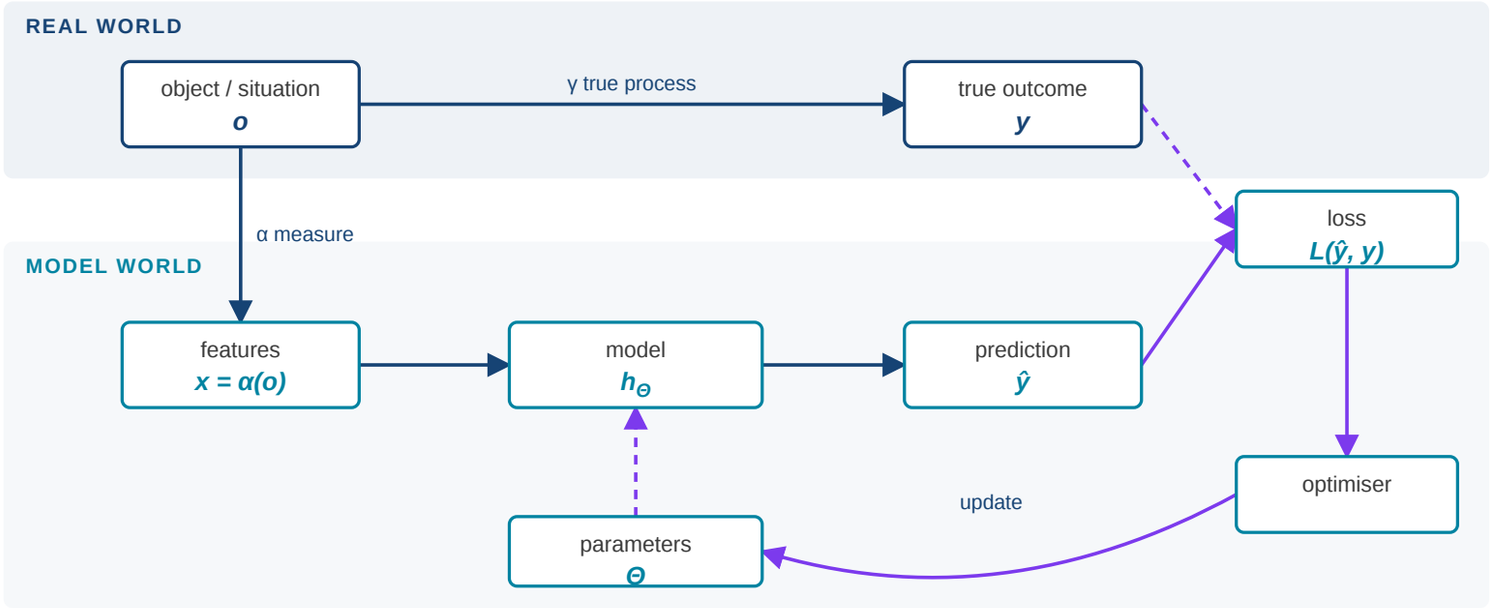
This is the consequence slide the terminology was for. It is also the bridge out: we now know uncertainty matters and comes in kinds — but we still have no systematic way to write it down, combine it or update it. That is where the lecture is heading.

Where Uncertainty Enters

Guiding Question: You have built learning systems before. Which of their parts were uncertain — and which of those did your model ever report?

The Learning System You Already Know

The optimiser changes the parameters — and the training loop closes



Which of these boxes did any model you have trained ever report its uncertainty about?

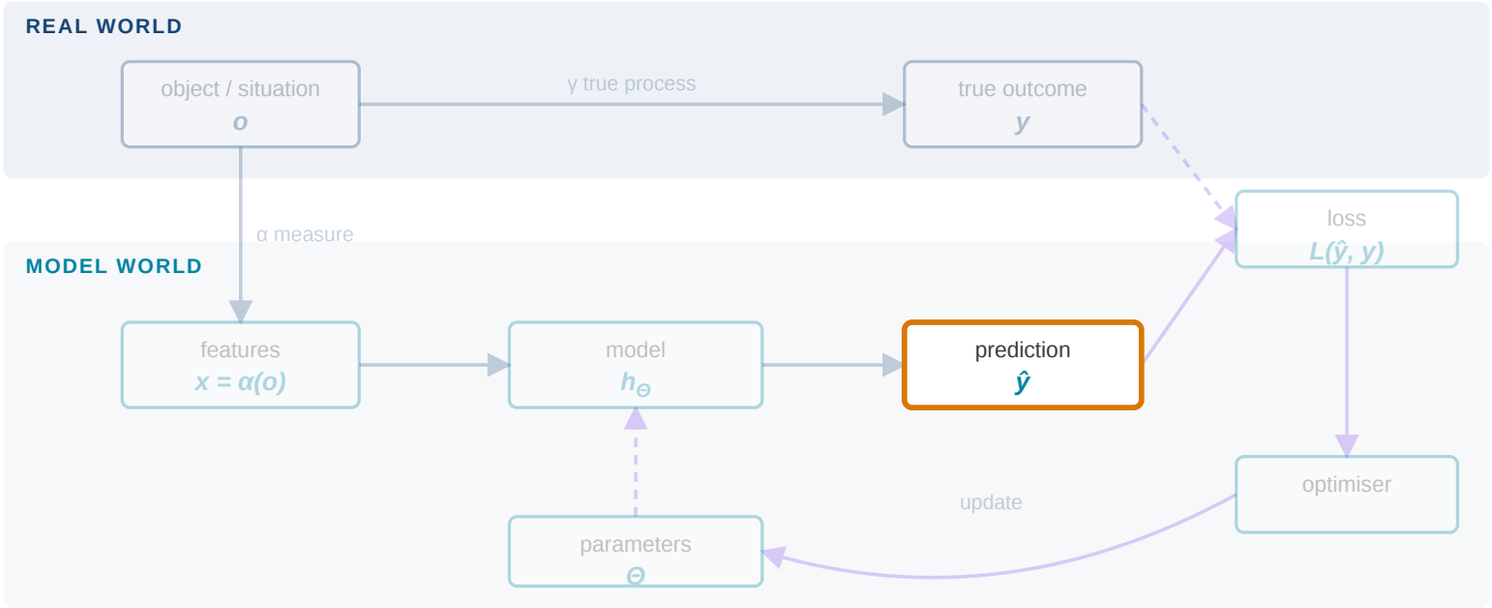
Speaker notes

A recap, not a lesson: the students have built such systems in earlier courses. Step through the four clicks quickly and name the boxes as a sentence — measure the situation, predict, compare with what happened, adjust the parameters, repeat. Data, model, loss, optimiser.

Then ask the question. The honest answer is usually "none, or only the final score" — which is exactly the gap the next slide opens. If the group has not seen the notation: γ is the real process, α the representation (what we measure), h_Θ the model with parameters Θ.

The Same System, Read for Uncertainty

In the evaluation



a finite test set, and a future that may not resemble the past

Speaker notes

The same figure the module has been building, now read a second way. One click per source, each highlighting the box it lives in:

1. **the world** — identical situations, different outcomes (aleatoric)
2. **the data** — noisy measurement, missing values, an unrepresentative sample
3. **the model** — the family may not contain the truth
4. **the parameters** — too few examples to pin them down (epistemic; this is the one more data reduces)
5. **the objective** — the loss is a *design choice*; it may not capture the task
6. **the optimisation** — stochastic steps, initialisation, approximate solutions
7. **the evaluation** — a finite test set, and a future that may differ

Ask before each click: where would *you* look next?

Two Distinctions Worth Keeping Straight

UNCERTAINTY *IN* THE LOSS

The quantities entering the loss are uncertain: y is noisy, the sample is finite. The loss itself is an ordinary function.

UNCERTAINTY *ABOUT* THE LOSS

Whether this objective represents the task at all. That is a modelling judgement, not a random variable.

> Do not say "the loss is random". Say what is uncertain: the data it is computed from, or our confidence that it is the right objective.

Speaker notes

The first pair matters because students conflate them and then write things that are false. The closing statement is the return to the course's opening claim, now with a system diagram behind it rather than a fraud example.

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Uncertainty is not a property of the final prediction. It **enters throughout the system** — and probabilistic machine learning is the language for representing and reasoning about it at every one of these places.

Beyond True or False

Guiding Question: A proof either holds or it does not. A learned prediction is never a proof — so what kind of conclusion is it, and how strong is it?

Three Modes of Reasoning

	From	To	What it guarantees
Deduction	premises	what **must** follow	certain — but adds no new content
Induction	observed patterns	a **regularity**, a hypothesis	nothing — content-adding, not guaranteed
Abduction	an observation	the **best available explanation**	nothing — only comparable against rivals

Machine learning lives in the bottom two rows.

Speaker notes

One criterion across the three rows: what does each mode guarantee, and what does it buy? Deduction guarantees everything and discovers nothing; induction and abduction discover and guarantee nothing. The rest of this unit shows what happens when they meet in practice — and why the choice between explanations is a decision, not a derivation.

The Cycle, on a Real Case: Uranus

DEDUCTION

Newton's laws plus the known planets give the orbit Uranus **must** follow.

INDUCTION

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

ABDUCTION

Best explanation: an **unseen planet** further out, pulling on Uranus.

Speaker notes

Uranus was discovered in 1781; by the 1840s its tables no longer matched the sky. The point of the slide is the last step: an explanation earns its keep by predicting something checkable — here, a position in the sky. The success is famous. The next slide asks why it was not obvious.

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ABDUCTION

Best explanation: an **unseen planet** further out, pulling on Uranus.

The cycle turns: the abduced planet becomes a new premise. Le Verrier deduces where it must be; in 1846 Galle points a telescope there and finds **Neptune**, about a degree from the prediction.

An Anomaly Leaves Two Moves

SAVE THE THEORY

Add a hidden cause

The bet: the theory is right; our picture of the world is incomplete.

If wrong: we chase something that is not there.

CHANGE THE THEORY

Replace the premises

The bet: the theory fails here, however well it worked elsewhere.

If wrong: we throw away two centuries of correct predictions.

1859: Mercury's orbit turns about **43″ per century** more than Newton allows. Which move would you make?

Speaker notes

Collect votes before the next slide. After Neptune, almost everyone chooses "save the theory" — and that is the right answer *for 1859*: the same scientist had just been spectacularly vindicated by exactly that move. Make students commit; the next slide pays off only if they have.

Same Move, Opposite Outcomes

anomaly	the move	what happened
Uranus, 1840s	a hidden planet	**Neptune** found, 1846 ✓
Mercury, 1859	a hidden planet, "Vulcan" (Le Verrier again)	?

Speaker notes

Same scientist, same method, same kind of anomaly, opposite outcomes. Vulcan was searched for at eclipses for decades and a few sightings were claimed; none survived. Einstein's theory produced the 43″ per century as a consequence, with nothing adjusted to fit it — which is why it replaced Newton rather than merely patching him.

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Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

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Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

> Both decisions were **reasonable when they were made**. What separated them was evidence that did not exist yet.

Deciding Before the Evidence Is In

- → **Strength of belief.** After two centuries of success, absorbing one anomaly *into* the theory was the rational first move.
- → **Evidence accumulates.** Every failed search made "a hidden planet" less plausible — belief moved step by step, not in one jump.
- → **Replace, don't just refute.** Newton fell when a rival explained *more*, not at the first failure.
- → **Costs decide.** Chasing Vulcan forever and abandoning a good theory too early are both expensive; which move to make depends on what being wrong costs.

> Beliefs of different strength, updated by evidence, acted on under costs — the whole lecture in one historical case. Lecture 02 gives each of these words its mathematics.

Speaker notes

Map each bullet to the course without naming the formulas: strength of belief is a prior, the accumulating searches are updating (Bayes, L02), "a rival explains more" is comparing models by how probable they make the data, and "costs decide" is the loan threshold from the start of the lecture.

Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

Speaker notes

The Uranus/Mercury fork in machine-learning form: collecting data is saving the theory (the model family is fine, the sample is incomplete); changing the features or the family is changing the theory. The three candidates are the sources of uncertainty from earlier in the lecture — noise (aleatoric), too little data (epistemic), and the model box of "Where Uncertainty Enters".

Mini-Exercise — Your Model Fails on a Subgroup

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Noise, too little data, or the wrong model?

Not too little data — then the error would have dropped. Left: the wrong model (try new features or a richer model) or noise (nothing helps, and experts with the same information cannot tell either). You saved the theory once; now the evidence says to change it.

Takeaway

> Learning from data never proves; it **bets**. An anomaly can be absorbed or can overturn the model, and which move is right is decided **under uncertainty** — by how strong the belief is, how the evidence accumulates, and what being wrong costs.

Beyond True or False

	Classical reasoning	Probabilistic reasoning
Conclusions	follow, or do not	are supported to *degrees*
Evidence	premises, taken as given	may be partial or noisy
Certainty	the conclusion is certain	the conclusion may stay uncertain
New information	can only add premises	can *change* what we believe

Classical logic asks *whether* a conclusion follows. Probabilistic reasoning also asks **how strongly the evidence supports each possible conclusion**.

Speaker notes

Connect to the three modes of reasoning just seen: deduction is the left card; induction and abduction live on the right, and they are the modes machine learning actually performs. A learned prediction is never a theorem — it is an inference from evidence, and its strength is the thing we want to write down.

What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

1. **Represent** — say how much we believe each possibility
2. **Combine** — put several uncertain pieces of information together
3. **Update** — change our beliefs when new evidence arrives
4. **Decide** — act on beliefs that are not certainties

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> **Probability provides that language.** For representing, combining and updating there is essentially one consistent set of rules; deciding adds a second ingredient — what the outcomes cost.

Takeaway

Machine-learning predictions are inferences from evidence, not derivations from premises. **Probability is the calculus of such inferences** — which is why the next lecture is about it, and why it will feel like an answer rather than a prerequisite.

Speaker notes

The reveal is the pivot of the whole introduction: the mathematics is now the solution to a problem the room has already felt. Say explicitly that "one consistent set of rules" is a real theorem (Cox), not a slogan — and that it covers beliefs only. Acting on them needs utilities as well (von Neumann–Morgenstern, Savage); the loan exercise already showed that: the threshold came from the costs. Leave it there.

What Probabilistic ML Enables

Guiding Question: Which questions about a machine-learning system will you be able to answer after this course that you cannot answer today?

After This Course, You Will Be Able to Ask — and Answer

ABOUT A PREDICTION

- ? How certain is it, and should I trust it?
- ? When should the model refuse to decide?
- ? What alternatives are plausible?

ABOUT THE DATA

- ? When would more data help?
- ? When is uncertainty unavoidable?
- ? What does this data actually justify?

ABOUT LEARNING

- ? How should belief change when evidence arrives?
- ? How do I combine prior knowledge with observations?
- ? How do I compare competing explanations?

ABOUT THE METHODS

- ? Why do the common methods work, and under which assumptions?
- ? When should I expect them to fail?
- ? What is my model assuming without telling me?

Speaker notes

Utility before curriculum structure. These are not learning objectives to be graded; they are the questions the students should recognise as *theirs* by the end. Ask which of them they can answer today — usually none with confidence.

What Changes Compared to Traditional Machine Learning

TRADITIONAL ML ASKS

What is the prediction?

PROBABILISTIC ML ALSO ASKS

- ? **How certain** is it?
- ? **What alternatives** are plausible?
- ? **Why** are we uncertain?
- ? **How does new evidence** change what we believe?
- ? **What decision** follows from this?
- ? **What can and cannot** be inferred from this information?

Takeaway

> Probabilistic machine learning does not replace prediction. It **surrounds every prediction with the questions a decision needs answered** — and gives one language in which all of them can be asked.

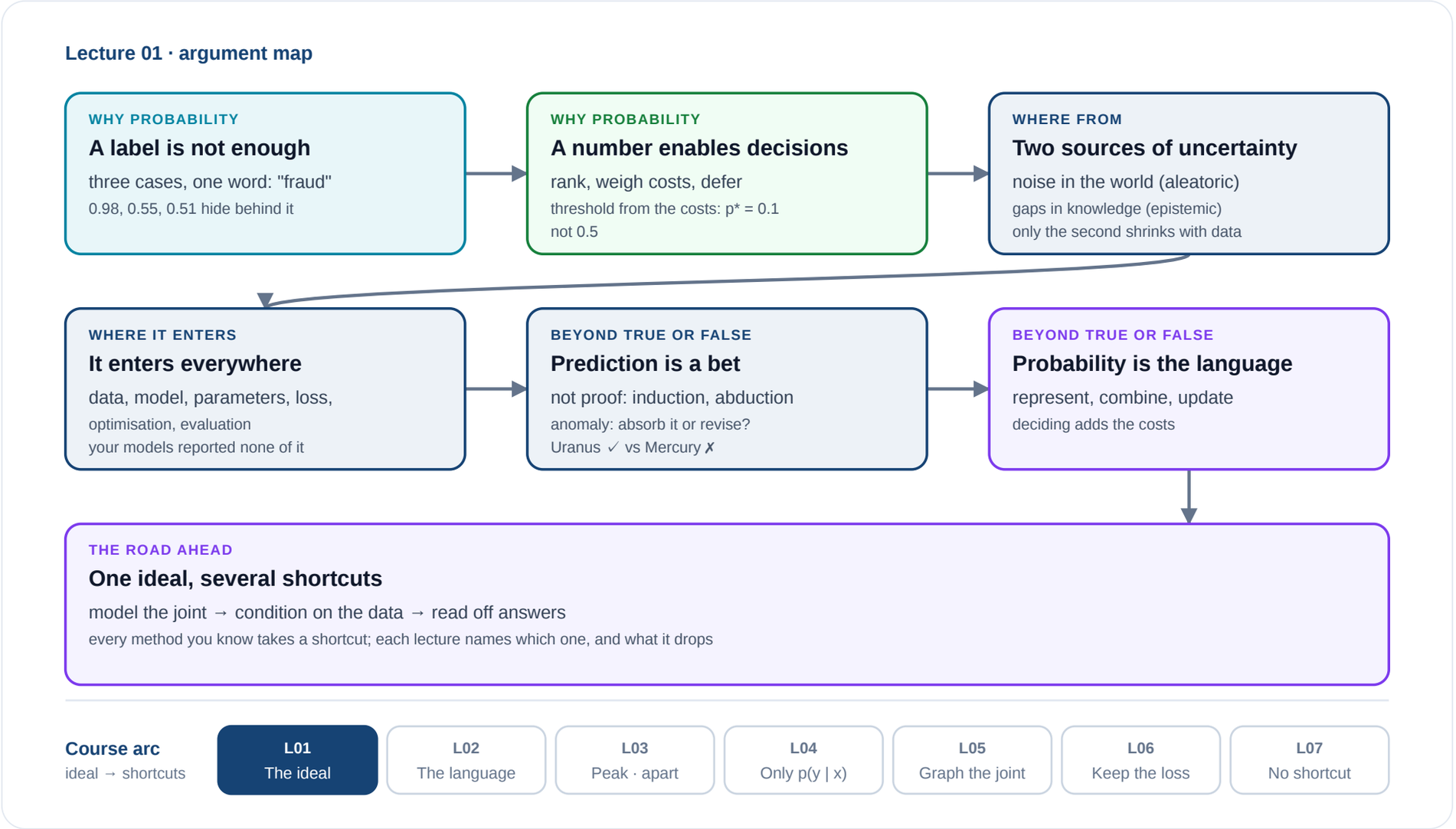
Speaker notes

Closing frame for the motivation half of the lecture. The right-hand card is the course in six questions; each later lecture answers one or two of them, which the roadmap at the end will make explicit.

The Road Ahead

Guiding Question: Now that the motivation is in place — which lecture answers which of your questions?

Argument Map



Speaker notes

Recap of the whole lecture, shown once the room has made every step. Read it top-left to bottom: a label hides what a number shows; the number enables decisions; uncertainty has two sources and enters every part of a learning system; a prediction is an inference, so it needs a language of degrees; probability is that language. The bottom strip is the course: this lecture states the ideal, each later lecture takes one shortcut, and Lecture 07 takes none. Every later lecture opens with its own map and the same strip.

The Road Ahead

Each lecture answers one of the questions you have just met.

02

Representing uncertainty *How do we describe uncertainty mathematically?*

Probability, distributions, and how to summarise data.

03

A classifier from probability *What does a classifier look like when it is built straight from Bayes — and what does it cost?*

Keeping the peak, the generative route and naive Bayes.

04

Methods you know, re-read *What do linear and logistic regression assume — and when is their output a probability?*

Least squares as a likelihood, logistic regression, calibration.

05

Structured models *How do we write down a large, structured set of beliefs?*

Graphical models: drawing the assumptions, fitting them, computing with them.

06

Judging models *What is the quantity we actually want to minimise — and what did we get instead?*

Evaluation and empirical risk minimisation.

07

Deciding and refining *How do we keep uncertainty all the way to the decision?*

Mixtures, Bayesian regression, and what comes after.

Speaker notes

Only now the detailed structure — the students can read each question as one of their own. Point at the arc: represent → classify → re-read → structure → judge → decide.

The Ideal, in Three Moves

Every method in this course is measured against the same three moves, in the language you are about to learn:

Model the joint

Write down what the world would have to be like to produce what we observed — the data *and* the unknowns together.

Condition on the data

Inference is conditioning: fix what was observed, and let the rules of probability say what follows for the rest.

Read off answers

Predictions, uncertainties and decisions are all quantities computed from that one conditioned model.

> Every one of these words — joint, condition, inference — gets a precise meaning in Lecture 02. They appear here only so that you recognise them when they return.

Speaker notes

This is the former "programme" slide, moved to the end where its vocabulary is motivated rather than a burden. Say "ideal" out loud: the next slide admits that most methods the students already know do not perform all three moves.

…and the Shortcuts Everyone Takes

Most methods you already know take a shortcut — and each one drops something.

Shortcut	Instead	What is lost	L
Keep the peak	the single most probable parameter (maximum likelihood)	how sure we are about it	03–04
Assume it apart	factorise the joint by independence	the dependences assumed away	03
Only what you need	model $p(y \mid x)$, never the joint (discriminative)	every other question	04
Graph the joint	state exactly which independences hold	cheap inference, in general	05
Keep the loss	minimise average error on the sample (ERM)	every probability	06
No shortcut	keep the whole posterior — EM only for the hidden labels	computation; it cannot see a wrong model family	07

> Knowing which shortcut a method takes tells you which questions it can no longer answer.

Speaker notes

This slide is the course's honesty clause. Students arrive knowing linear and logistic regression, trees, maybe neural networks — all of them shortcuts. The point is not that shortcuts are bad; it is that each one silently drops a kind of answer, and the course teaches them to name which.

Return to this table at the start of every later lecture: each lecture's argument map has the same strip at the bottom, with its own row highlighted. The generative/discriminative distinction is named here for the first time; Lecture 03 practises the generative side (naive Bayes) and Lecture 04 the discriminative side (logistic regression), and makes the comparison explicit.